Voltscope

Stop overpaying for energy. Start checking every bill.

Automated extraction, validation and intelligence for UK electricity and gas bills — the bill-intelligence module of the VoltEdge family.

AI Extraction · Deterministic Validation · Renewal Alerts · Duplicate-Charge Detection · Portfolio Insights · JSON / Excel / CSV Export · REST API

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01 THE PROBLEM WE SOLVE

UK businesses and households are routinely overbilled on energy, and almost nobody checks. Bills arrive as PDFs and phone photos in dozens of supplier-specific layouts, burying the figures that matter — MPAN, unit rates, standing charges, meter reads and contract dates. Estimated reads overcharge, duplicate line items slip through, totals quietly fail to add up, and fixed contracts roll onto expensive out-of-contract rates because renewal dates go unnoticed. Checking this by hand across a portfolio is slow and error-prone. Voltscope reads each bill, validates every figure, and flags exactly what is wrong — deterministically, in seconds.

50	10	7	1	0
AUTOMATED TESTS PASSING	DETERMINISTIC VALIDATION CHECKS	BILL FORMATS ACCEPTED	API CALL PER BILL (CENTS EACH)	LLM CALLS IN VALIDATION & ANALYTICS

02 KEY FEATURES

Structured extraction — Every field from any UK electricity or gas bill: supplier, MPAN/MPRN, tariffs, unit rates, standing charges, meter reads, charges and totals.

Human-in-the-loop review — Web interface to review extracted fields beside findings, correct anything, and re-validate instantly on save.

Multi-page & HEIC — Combine several photos into one bill; iPhone HEIC handled automatically alongside PDF, PNG, JPG, WEBP and GIF.

Export — JSON, a four-sheet Excel workbook and CSV for auditors, accountants and spreadsheets.

Deterministic validation — Ten registered checks, each finding carrying a severity, category and a plain-English recommendation.

Portfolio intelligence — Renewal alerts, estimated-read reports, duplicate-charge detection with quantified overcharge, spend summaries and supplier comparison.

High-unit-rate flags — Tariffs above a portfolio-relative baseline surfaced automatically, adapting as the portfolio grows.

REST API & audit trail — Full OpenAPI/Swagger docs, API-key auth, SQLite persistence and processing history.

03 TECHNICAL ARCHITECTURE

Extraction — Anthropic Claude API · vision over PDF & images · conservative JSON schema

Validation — Registry of deterministic checks · typed Finding model

Data layer — SQLite · lossless JSON + queryable columns · history

Export — openpyxl (Excel) · CSV · JSON

Testing — 50 tests · golden fixtures · pytest

Backend — FastAPI · Uvicorn · Pydantic v2

Intelligence — Pure-Python analytics · zero LLM calls

Frontend — Dependency-free vanilla JavaScript · single file

Auth — API-key (X-API-Key) · OpenAPI security scheme

Deployment — Docker · Render blueprint · $PORT-ready

04 WHO IT'S FOR

Energy consultancies & brokers — audit client bills at scale, surface overcharges, white-label via the API.

SMEs — catch billing errors without a dedicated energy analyst.

Property portfolios — hotels, retail chains and student accommodation.

Multi-site commercial operators — portfolios of bills with renewal tracking and spend comparison.

Bookkeepers & accountants — validated energy data straight into the ledger.

VoltEdge integration — feeds validated bill data into the wider VoltEdge energy-intelligence platform.

05 COMMERCIAL MODEL

Self-serve SaaS	API licence	White-label / OEM	Enterprise
£29–£99 per user / month SMEs & bookkeepers	**Per-call** or flat rate Integrations & consultancies	**Custom** per client tier Brokers auditing 50–200 clients	**Custom SLA** annual Multi-site operators & audit needs

Illustrative pricing for a productised version; the current build is proprietary and source-visible for evaluation.

06 FUTURE ROADMAP

Automatic multi-page detection — group a bill's pages without manual selection.

Historical trends — track a site's rates and consumption over time.

Accounting integrations — Xero, QuickBooks and Sage export.

Broader bill types — water, telecoms and waste beyond energy.

Anomaly baselines — flag bills that deviate from a site's own history.

Scale-out — PostgreSQL and multi-tenant isolation beyond single-instance SQLite.

PORTFOLIO
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GITHUB
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DEPLOY
Docker · Render-ready